

HYDRAULIC AND PNEUMATIC STORED ENERGY

Manage sudden movement or release from pressurized fluids, accumulators, cylinders, hoses and raised components through isolation, depressurization, mechanical blocking and zero-energy verification.

PURPOSE AND SCOPE

Manage sudden movement or release from pressurized fluids, accumulators, cylinders, hoses and raised components through isolation, depressurization, mechanical blocking and zero-energy verification.

KEY HAZARDS

1  TRAPPED PRESSURE	2  INJECTION INJURY	3  HOSE WHIP	4  DROPPING EQUIPMENT
5  ACCUMULATOR	6  UNEXPECTED MOVEMENT	7  HOT FLUID	8  POOR BLOCKING

CRITICAL VALUES TO VERIFY IN THE FIELD

0 bar Verified pressure before intervention	100% Mechanical blocking for raised loads	0 Tolerance for hand-checking leaks
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REMEMBER - SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

HYDRAULIC AND PNEUMATIC STORED ENERGY

Manage sudden movement or release from pressurized fluids, accumulators, cylinders, hoses and raised components through isolation, blocking and bleeding.

STANDARDS AND FIELD REQUIREMENTS

OSHA 1910.147, 1910.242 and manufacturer energy-release procedures require pressurized energy to be isolated, restrained, released and verified.

OSHA 1910.147(d)

ENERGY-CONTROL STEPS

Shut down, isolate every source, lock, release/restrain stored energy and verify isolation.

OSHA 1910.147(d)(5)

STORED ENERGY

Reduce pressure, gravity, spring and accumulator energy to a safe level and monitor reaccumulation.

OSHA 1910.242(a)

EQUIPMENT CONDITION

Keep hoses, fittings, valves and gauges suitable for service and pressure and in safe condition.

ÜRETİCİ PROSEDÜRÜ

BLOCKING AND BLEEDING

Support raised parts with approved mechanical devices and bleed accumulators and pilot lines at designated points.

MINIMUM EVIDENCE BEFORE START

- | | |
|--|--|
| ☐ Energy-control plan | ☐ Hydraulic/pneumatic schematic |
| ☐ LOTO record | ☐ Blocking-device inspection |
| ☐ Pressure-release verification | ☐ Hose/fitting maintenance record |

SUPERVISOR DISCUSSION POINTS

- Where can pressure remain trapped?
- Is the load supported only by a valve?
- How did we verify zero by a second method?

LOTO, MECHANICAL BLOCKING AND ZERO PRESSURE



Source isolation - Valve lock - Accumulator - Block - Gauge - Test point

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CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. ELIMINATE LIVE INTERVENTION

Design service for zero pressure with test ports and permanent blocking points.

2. ISOLATE AND MECHANICALLY SUPPORT

Isolate pump/compressor, lock valves and mechanically support raised parts.

3. BLEED AND VERIFY ZERO

Bleed accumulators and both sides of cylinders; verify zero at gauge and controlled test point.

4. REMOTE LEAK DETECTION

Detect leaks remotely with card/suitable tools and depressurize before approaching damaged hoses.

CRITICAL WARNING



A pinhole hydraulic leak can inject fluid deep into tissue. Never search by hand; stop work and obtain urgent surgical assessment after injury.

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|--|---|
| ☐ Energy sources identified | ☐ Pump/compressor locked out |
| ☐ Valves locked | ☐ Accumulator discharged |
| ☐ Load mechanically blocked | ☐ Zero pressure verified |
| ☐ Reaccumulation monitored | ☐ Hose/fitting sound |
| ☐ Leak test remote | |



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ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
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